Intelligent Embedded Systems

Cats AI — current status

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Project Overview

- Goal: analyze complex traffic scenes from video using multimodal AI models.
- Focus is not only event detection, but structured, human-aligned scene understanding.
- The model should identify relevant actors, visible risks, uncertainty, and scene complexity.
- Long-term direction: improve explanations through supervised fine-tuning (SFT), preference learning, and human-aligned feedback (DPO).

Current Progress

- Identified a suitable crash-video dataset with normal and accident clips.
- Built a Qwen-VL video querying pipeline for structured accident analysis.
- Designed JSON prompts and validation schema for consistent model outputs.
- Ran first evaluation experiments measuring valid JSON rate and crash-detection accuracy.
- Prepared the basic SFT data format and LoRA training pipeline.

Initial Experiment

- Task: classify whether a dashcam video contains a visible accident.
- Input: sampled video frames from normal and crash videos.
- Output: structured JSON with accident prediction.
- Metrics:
  - Valid JSON outputs: **100%**
  - Detection accuracy: **92.7%**
- Result: the pipeline works end-to-end and can be used for larger-scale experiments.

Possible Training Directions

1. **Dataset with analysis labels**
   Find a dataset that already contains accident explanations, risk factors, or scene-level reasoning labels.
2. **Knowledge distillation**
   Use a stronger teacher model to generate structured assistant JSONs, then fine-tune the local model on these outputs.
3. **Hybrid approach**
   Use datasets with partial analysis labels and let the teacher model enrich or complete them.

Chosen Direction

- For now, we choose **knowledge distillation.**
- Reason: no suitable dataset with the required analysis-style labels has been found yet.
- The teacher model will generate structured JSON outputs:
  - accident presence
  - confidence
  - scene description
  - main objects
  - risk factors
  - reasoning
  - uncertainty

Next Steps

- Generate assistant JSONs from a stronger teacher model.
- Validate generated JSONs against the schema.
- Filter low-quality or incorrect outputs using the existing crash/normal labels.
- Build the distilled SFT dataset.
- Run a small LoRA SFT experiment and compare before vs. after performance.